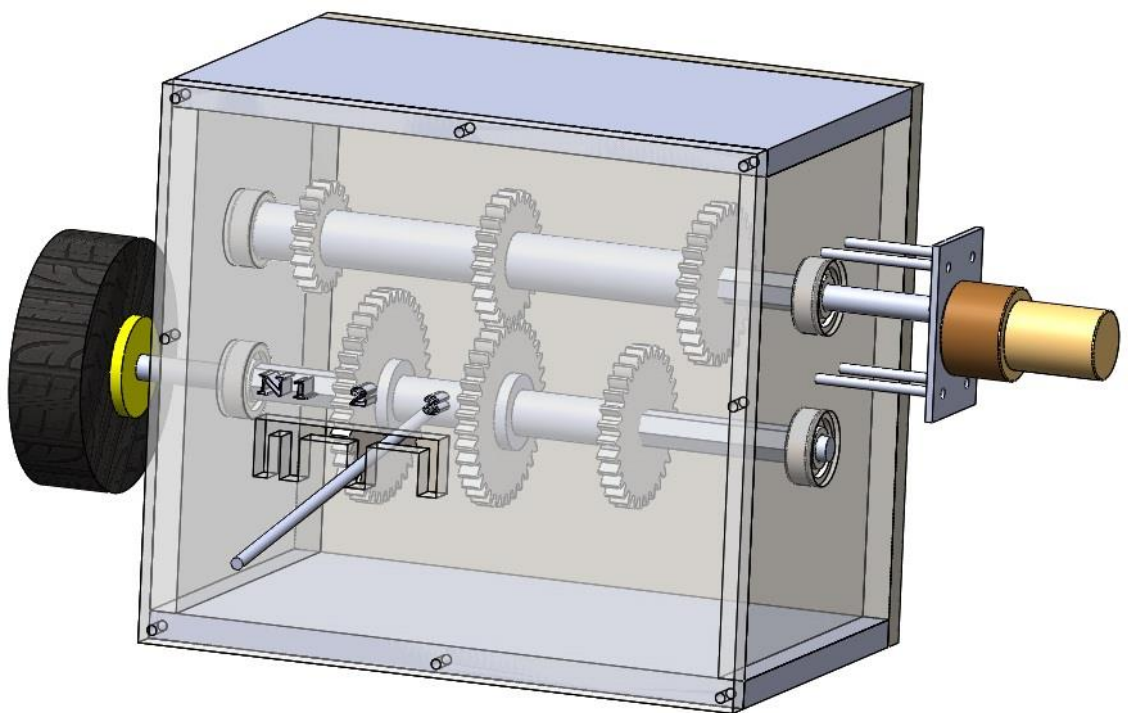


3 Speed Gearbox Mechanism

- Project Proposal

Dr.Eng. Ahmed Sameh



● Idea of the project

The 3 Speed Gearbox presents the mechanism of a gearbox. **Gears of various dimensions** are placed along two different shafts. A **motor** rotates the central shaft and different arrangements that can be made in the project causes the **rotational motion** to transfer from one shaft to another. This project can be made from **Mdf, mild steel** and **Acrylic**.

● How will it work?

The machine can act in 4 different **scenarios**:

○ Scenario 1

- When the shaft that determines the main movement is at “N” the wheel doesn’t move.

○ Scenario 2

- When the shaft is at “1”, the wheel moves at a certain speed.

○ Scenario 3

- When the shaft is at “2”, the wheel moves faster than its speed at Scenario 2.

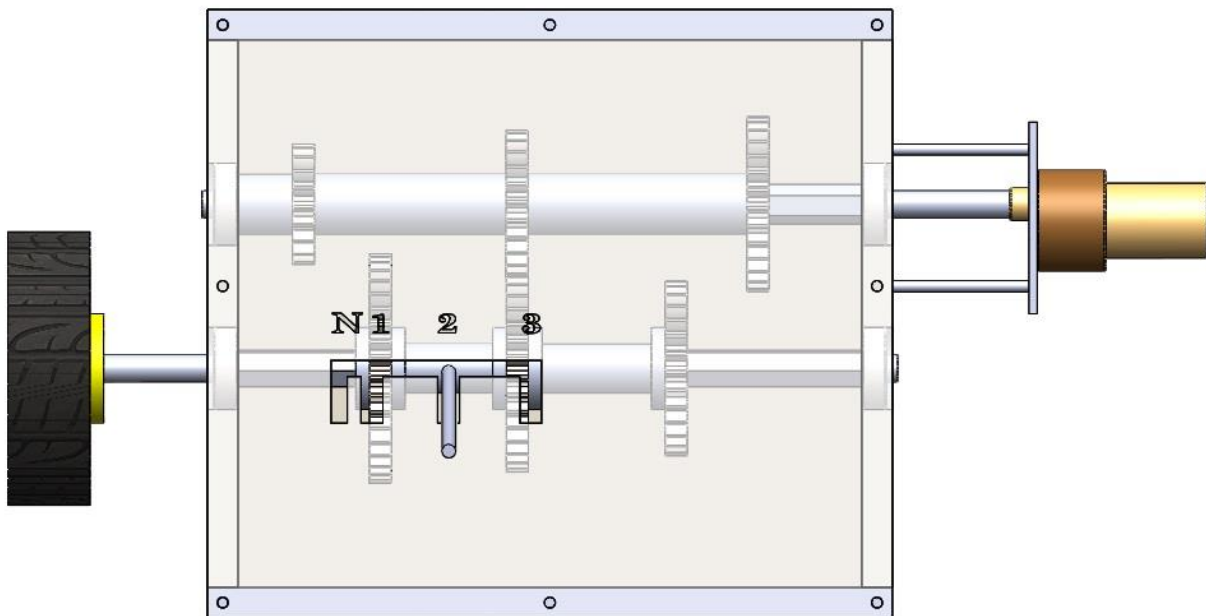
○ Scenario 4

- When the shaft is at “3”, the wheel moves faster than its speed at Scenario 3.

● Components

- Gears
- Couplers
- Bearings
- DC Motor
- Shafts
- Clamps & Mounts
- Wheel
- Base Frame
- Supporting Frame

● Side View



● Team

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